AI.111

Neural Networks

COURSE OVERVIEW

Artificial neural networks: perceptrons, multilayer networks, backpropagation, and training deep models.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: BRAINS IN SILICON

KEY TOPICS

- Biological inspiration
- The perceptron
- Multilayer perceptrons
- Neural networks in AI

TEACHING FOCUS

Introduce the neuron as a computational unit.

KEY TAKEAWAY Networks of simple units compute complex functions.

2

SESSION

60 MINUTES

ACTIVATION & FORWARD PASS

KEY TOPICS

- Sigmoid, tanh, ReLU activations
- Forward propagation
- Loss functions
- Output layers for regression/classification

TEACHING FOCUS

Compute predictions from inputs through layers.

KEY TAKEAWAY The forward pass turns data into predictions.

3

SESSION

60 MINUTES

BACKPROPAGATION

KEY TOPICS

- Chain rule and gradients
- Computing gradients backward
- Weight updates
- Stochastic gradient descent

TEACHING FOCUS

Train networks by propagating error.

KEY TAKEAWAY Backpropagation is how networks learn from mistakes.

4

SESSION

60 MINUTES

TRAINING DYNAMICS

KEY TOPICS

- Learning rate and momentum
- Batch vs. mini-batch
- Overfitting and regularization
- Dropout and early stopping

TEACHING FOCUS

Train models that generalize.

KEY TAKEAWAY Training well is harder than training fast.

5

SESSION

60 MINUTES

NETWORK ARCHITECTURES

KEY TOPICS

- Feedforward networks
- Convolutional networks intro
- Recurrent networks intro
- Autoencoders

TEACHING FOCUS

Match architecture to data type.

KEY TAKEAWAY Architecture encodes inductive bias.

6

SESSION

60 MINUTES

NEURAL NETWORKS IN ROBOTICS & REVIEW

KEY TOPICS

- Perception and control networks
- Deployment on embedded
- Course recap and assessment
- Toward deep learning (AI.210)

TEACHING FOCUS

Deploy learned models on robots.

KEY TAKEAWAY Learned models give robots adaptive behavior.